

**STATISTICAL ANALYSIS OF INFANT MORTALITY:
A TIME SERIES AND CLUSTER APPROACH**

PROJECT REPORT

By

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UNDER THE GUIDANCE OF

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DECLARATION

I hereby declare that the project entitled “STATISTICAL ANALYSIS OF INFANT MORTALITY:A TIME SERIES AND CLUSTER APPROACH” was carried out by me during the final semester of Master of Science in Statistics during the year 2024–26 in the Department of Studies in Statistics, University of Mysore, Manasagangothri.

Date:18/06/2026

Place: Mysuru

Snehagopinath C

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I also thank my parents, friends, and well-wishers for their motivation and encouragement throughout the project.

Finally, I thank everyone who directly or indirectly contributed to the successful completion of this work.

CERTIFICATE

This is to certify that the project entitled “STATISTICAL ANALYSIS OF INFANT MORTALITY:A TIME SERIES AND CLUSTER APPROACH” has been submitted by Miss Snehagopinath C and is the result of work carried out by her during the final semester of Master of Science in Statistics 2024–26, under my supervision in the Department of Studies in Statistics, University of Mysore, Manasagangothri, Mysuru - 570006.

Date: 18/06/2026

Place: Mysuru

Praveena A.S

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I: TIME SERIES ANALYSIS

India's Infant Mortality Rate — 1971 to 2024

1. Introduction

Infant Mortality Rate (IMR) is one of the important indicators used to measure the health status and socio-economic development of a country. It is defined as the number of infant deaths below one year of age per 1,000 live births in a given year. It reflects the quality of healthcare services, maternal health, nutrition, sanitation, and overall living conditions of a population.

In India, infant mortality has been an important public health issue for many years. A high infant mortality rate indicates poor healthcare facilities, lack of proper nutrition, inadequate maternal care, and poor sanitation conditions. On the other hand, a low infant mortality rate indicates better healthcare services and improved socio-economic conditions. India has made considerable progress in reducing infant mortality through various government initiatives such as immunization programs, maternal and child healthcare schemes, improved hospital facilities, and awareness campaigns. These efforts have contributed to a gradual decline in infant deaths over the years. However, significant regional differences still exist, with some states showing higher infant mortality rates due to differences in healthcare access, education, and economic conditions.

The study of infant mortality rate is essential for understanding child health conditions and evaluating the effectiveness of public health policies. Analyzing infant mortality trends in India helps identify patterns, measure progress, and suggest appropriate policy measures for further improvement in child survival and welfare.

2. Objectives

- To analyse the trend in India's Infant Mortality Rate from 1971 to 2024..
- To identify the optimal ARIMA model.
- To validate the fitted model.
- To forecast India's IMR for the next 10 years using the fitted ARIMA model.

3. Description of the Data

The time series dataset contains annual Infant Mortality Rate (IMR) observations for India from 1971 to 2024, comprising 54 observations. The IMR is measured as the number of deaths of infants under one year of age per 1,000 live births.

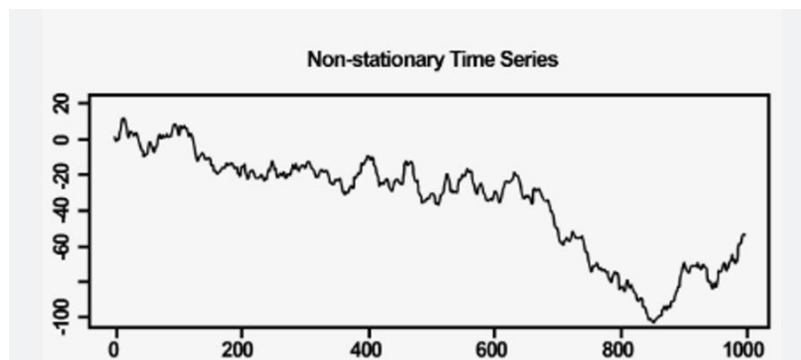
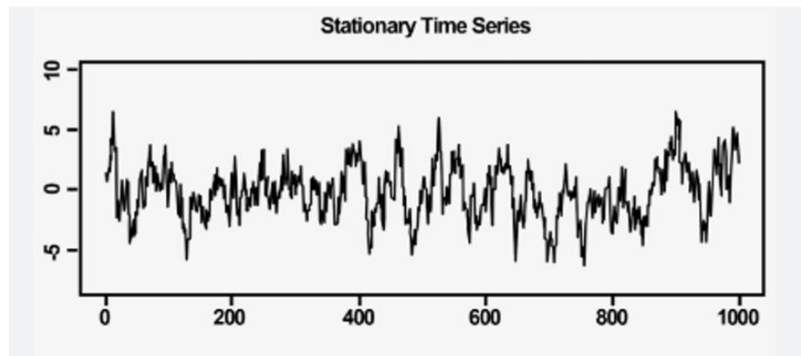
Attribute	Details	Value
Country	India	—
Time Period	1971 – 2024	54 years
Minimum IMR	Lowest recorded	23
Maximum IMR	Highest recorded	133
Mean IMR	Average over all years	72.57
Trend	Consistent decline	↓ Downward

- Source: World Bank Open Data —
<https://data.worldbank.org/indicator/SP.DYN.IMRT.IN?locations=IN>

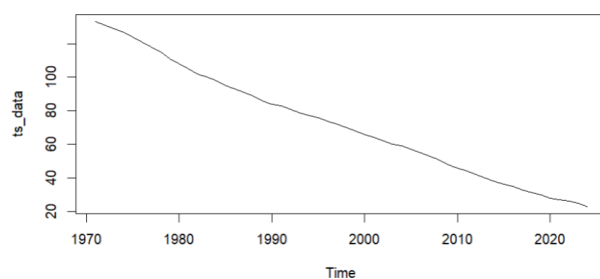
4. Analysis and Interpretation

What is Stationary and Non-Stationary Data?

A time series is said to be stationary if its statistical properties — mean and variance — remain constant over time with no trend or seasonality. If any of these conditions is violated, the series is non-stationary and must be transformed before modelling.



4.1 Time Series Plot



The time series plot of India's IMR from 1971 to 2024 shows a strong, steady declining trend over the entire period. No seasonal pattern is observed, and the variance is approximately constant. The clear downward trend indicates non-stationarity in the mean.

4.2 ADF Test — Augmented Dickey-Fuller Test

The Augmented Dickey-Fuller (ADF) test is a common statistical test used to determine whether a given time series is stationary or not.

Hypothesis:

H_0 : The time series is non-stationary (unit root exists).

H_1 : The time series is stationary.

Result: The p-value obtained is 0.08767, which is greater than 0.05. Therefore, we do not reject the null hypothesis at 5% level of significance and conclude that the data is non-stationary.

4.3 KPSS Test

The KPSS test is another approach for checking stationarity. Its hypotheses are opposite to the ADF test:

H_0 : The process is stationary.

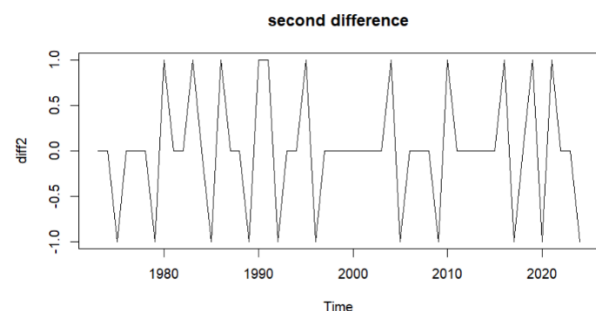
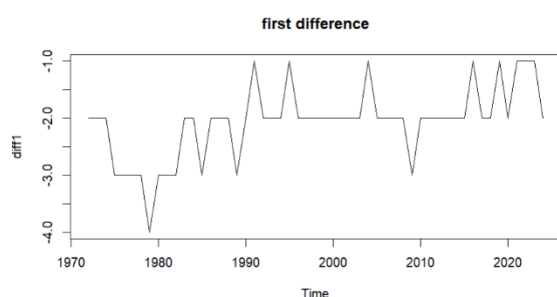
H_1 : The series is non-stationary.

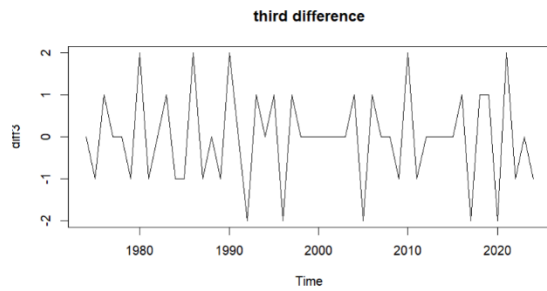
Result: The p-value is 0.01, which is less than 0.05. Therefore, we reject the null hypothesis at 5% level of significance and conclude that the data is non-stationary.

4.4 Differencing

Non-stationary data can be made stationary through differencing. A logarithmic transformation is a common technique used to stabilise the variance in time series data. The formula applied is:

$$y'_i = \ln(y_i)$$





After applying differencing, the series became stationary at the second difference. Therefore, the appropriate order of differencing is $d = 2$. Both ADF and KPSS tests confirmed stationarity after second differencing.

4.5 ARIMA Model — ARIMA(p, d, q)

The ARIMA model (Auto Regressive Integrated Moving Average) creates a linear equation which describes and forecasts time series data. Its three components are:

Autoregression (AR): Refers to a model that shows a changing variable that regresses on its own lagged, or prior, values.

Integrated (I): Represents the differencing of raw observations to allow the time series to become stationary.

Moving Average (MA): Incorporates the dependency between an observation and a residual error from a moving average model applied to lagged observations.

ARIMA Parameters

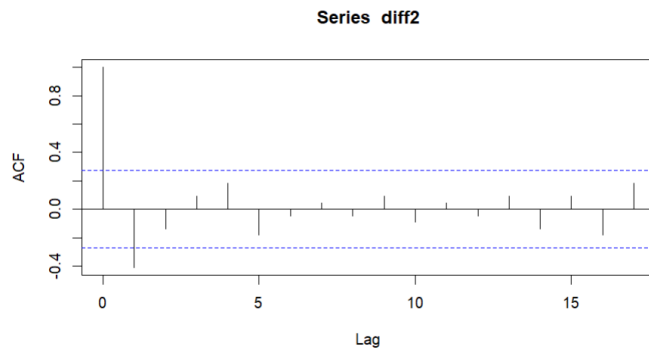
The standard notation for ARIMA is ARIMA(p, d, q), where:

- p: The number of lag observations (order of Autoregressive model).
- d: The number of times the raw observations are differenced (degree of differencing).
- q: The size of the moving average window (order of the moving average).

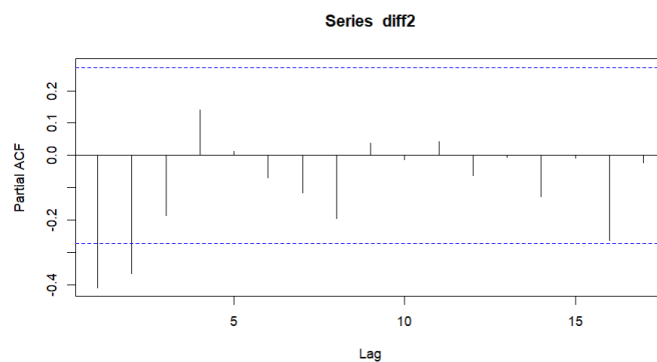
4.6 ACF and PACF

The Autocorrelation Function (ACF) plot shows the autocorrelation between the time series and its lagged versions. It is used to determine the order of the Moving Average (MA) process. The Partial Autocorrelation Function (PACF) plot shows the partial autocorrelation controlling for shorter lags. It is used to determine the order of the Autoregressive (AR) process.

ACF PLOT



PACF PLOT



From the ACF and PACF plots, AR(2) and MA(1) were identified as the most suitable initial choices. Since the second difference was applied, $d = 2$ is fixed.

4.7 AIC and BIC — Model Selection

AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) are statistical measures used for model comparison. A lower value indicates a better-fitting model.

AIC Formula: $AIC = -2 \times \log\text{-likelihood} + 2k$

BIC Formula: $BIC = -2 \times \log\text{-likelihood} + k \times \log(n)$

The competing models considered were: ARIMA(2,2,1), ARIMA(3,2,1), ARIMA(4,2,1), ARIMA(3,2,0), ARIMA(2,2,2). Both AIC and BIC gave their minimum values for ARIMA(2,2,2), which was therefore selected as the most appropriate model.

Model	AIC	BIC	Selected
ARIMA(2,2,1)	Higher	Higher	No
ARIMA(3,2,1)	Higher	Higher	No
ARIMA(4,2,1)	Higher	Higher	No
ARIMA(3,2,0)	Higher	Higher	No
ARIMA(2,2,2)	Minimum	Minimum	✓ Yes

4.8 Model Fitting — ARIMA(2,2,2)

The ARIMA(2,2,2) model was fitted to the log-transformed IMR data. The model produced statistically significant coefficient estimates and a good fit to the observed data.

The fitted arima Model is:

$$Y_t = 0.1756Y_{t-1} - 0.5552Y_{t-2} - 0.8486e_{t-1} + 0.7363e_{t-2}$$

4.9 Ljung-Box Test

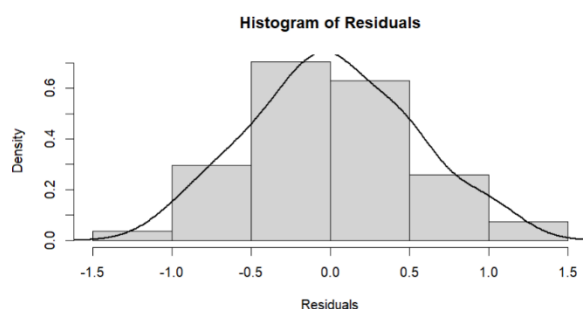
The Ljung-Box test checks whether autocorrelation exists in the residuals of the fitted model.

H₀: Residuals are independently distributed (no autocorrelation).

H₁: Residuals are not independently distributed (autocorrelation exists).

Result: The p-value is 0.922, which is greater than 0.05. Therefore, we fail to reject the null hypothesis and conclude that the residuals are independently distributed. The histogram of residuals confirms that they are approximately normally distributed

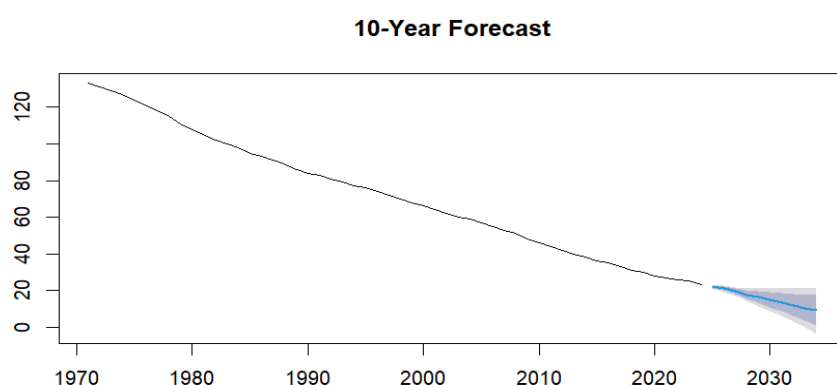
HISTOGRAM OF RESIDUALS



Here we can see that the residuals are normally distributed

4.10 Forecasting — Next 10 Years

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	21.964357	21.3083116	22.62040	20.961022	22.96769
2026	20.891829	19.8017319	21.98193	19.224670	22.55899
2027	19.277366	17.7098763	20.84486	16.880098	21.67463
2028	17.588236	15.3235563	19.85292	14.124707	21.05176
2029	16.186905	13.0544537	19.31936	11.396234	20.97758
2030	14.877562	10.8337723	18.92135	8.693120	21.06200
2031	13.424569	8.4398138	18.40932	5.801044	21.04809
2032	11.895279	5.8941505	17.89641	2.717345	21.07321
2033	10.432356	3.3290675	17.53564	-0.431186	21.29590
2034	9.023447	0.7579128	17.28898	-3.617597	21.66449



Using the fitted ARIMA(2,2,2) model, India's IMR was forecasted for 10 years beyond 2024. The forecasted values show a continued statistically consistent declining trend, indicating sustained improvement in child survival outcomes, contingent on maintained policy efforts.

II: CLUSTER ANALYSIS

Infant Mortality Rate Across 149 Countries — 2020 to 2024

5. Introduction

While Part I examined the temporal dynamics of India's IMR through time series analysis, Part II adopts a cross-sectional and comparative perspective. Cluster analysis is applied to IMR data from 149 countries over the period 2020 to 2024, with the aim of identifying natural groupings of countries that share similar levels of infant mortality.

Cluster analysis is a multivariate unsupervised statistical technique that groups objects or observations into clusters such that those within the same group are more similar to each other than to those in other groups. Applied to multi-country IMR data, it enables the identification of global patterns in child health outcomes, which is valuable for international policy comparison and resource allocation.

The dataset used in this part was sourced from the World Bank Open Data portal and contains IMR values for 158 countries across five years: 2020, 2021, 2022, 2023, and 2024. These five annual observations per country serve as the multivariate input for clustering.

Three approaches are applied: the scree plot is used to determine the optimal number of clusters, Hierarchical Clustering using Ward's linkage to visualise the dendrogram, and k-Means Clustering to obtain final, quantitative cluster assignments.

6. Objectives

- To group 149 countries into clusters based on their IMR values across 2020–2024.
- To determine the optimal number of clusters using the Elbow Method (Within-Group Sum of Squares).
- To perform Hierarchical Clustering using Ward's linkage and visualise results via a dendrogram.
- To apply k-Means Clustering and obtain final cluster assignments with centroid values.
- To interpret the characteristics of each cluster in terms of IMR levels and geographic/development patterns.
- To identify which countries share similar infant mortality profiles and what that implies for global health policy.

7. Description of the Data:

The dataset contains Infant Mortality Rate (IMR) observations for 149 countries for the years 2020, 2021, 2022, 2023, and 2024. Each country has five IMR values (one per year), which together form the multivariate input for clustering. The IMR is measured as deaths per 1,000 live births.

Attribute	Details	Value
Number of Countries	Total observations	149
Time Period	Years covered	2020 – 2024
Variables per Country	Annual IMR values	5 (one per year)
Lowest IMR (2024)	San Marino	1.2
Highest IMR (2024)	Central African Rep.	58.8

Source: World Bank Open Data:

<https://data.worldbank.org/indicator/SP.DYN.IMRT.IN?locations=IN>

8. Methodology

8.1 Scree Plot

The Within-Group Sum of Squares (WSS) is computed for $k = 1$ to 10. The WSS decreases as k increases; the optimal k is identified at the 'elbow' — the point where the rate of decrease slows sharply. The formula is:

$$WSS = \sum_i \sum_{i \in C_i} ||x_i - \mu_i||^2$$

8.2 Hierarchical Clustering (Ward's Method)

Hierarchical clustering builds a dendrogram by iteratively merging the two closest clusters. Ward's linkage minimises the total within-cluster variance at each step. The distance matrix is computed using Euclidean distance across the 5-year IMR values:

$$d(x, y) = \sqrt{\sum_k (x_k - y_k)^2} \text{ where } k = 2020, 2021, 2022, 2023, 2024$$

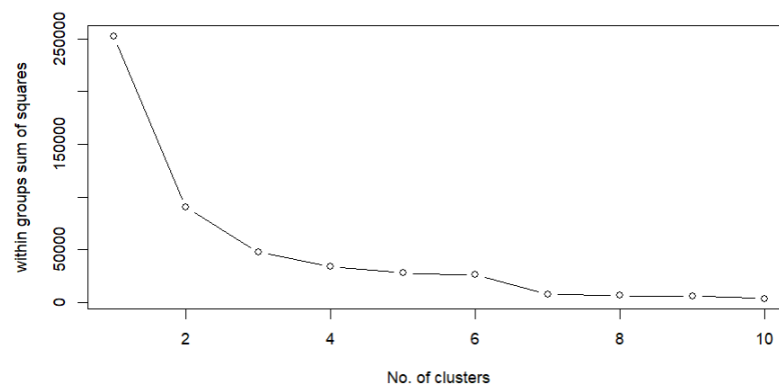
The dendrogram was cut at $k = 3$ clusters, confirmed by the Elbow Method. Country names were used as labels.

8.3 K-Means Clustering

K-Means partitions the 149 countries into $k = 3$ clusters by minimising the within-cluster sum of squares. The algorithm iterates between assigning countries to the nearest centroid and recomputing centroids until convergence. The aggregate function was used to compute the mean IMR of each cluster per year.

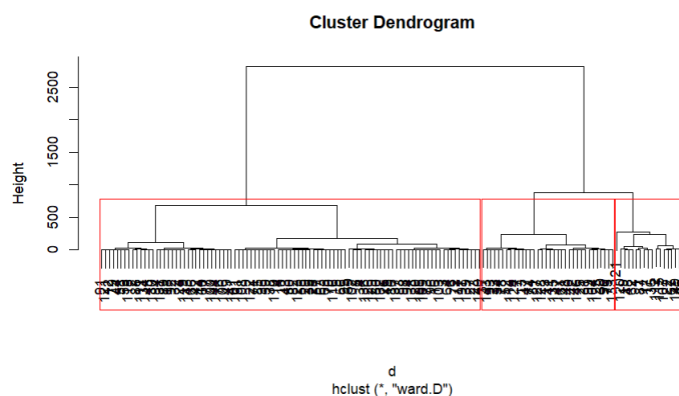
9. Analysis and Results

Scree Plot



The plot showed a clear elbow at $k = 3$, confirming three clusters as the optimal grouping for the 149 countries. Beyond $k = 3$, additional clusters provide diminishing reductions in within-group variance.

Hierarchical Clustering



The dendrogram using Ward's linkage clearly reveals three well-separated groups of countries. The high merge height at the point of cutting into 3 clusters indicates strong inter-cluster dissimilarity and high intra-cluster cohesion. Countries with very similar IMR profiles across 2020–2024 are merged early (low in the dendrogram), while the three broad groupings are separated by large heights.

k-Means Clustering Results

K-Means clustering with $k = 3$ converged to a stable solution. The three clusters are characterised by their mean IMR values across 2020–2024:

Cluster-wise Mean Values (2020–2024)					
K-Means Clustering Output ($k = 3$)					
Cluster	2020	2021	2022	2023	2024
Cluster 1 (High)	63.89	65.03	76.05	59.95	58.65
Cluster 2 (Moderate)	32.43	31.76	31.07	30.46	29.68
Cluster 3 (Low)	8.34	8.17	8.01	7.99	7.71

 Cluster 1 — High IMR (IMR per 1,000 live births)

#	Country
1	Afghanistan
2	Benin
3	Central African Republic
4	Chad
5	Guinea
6	Lesotho
7	Liberia
8	Pakistan
9	Serbia
10	Slovak Republic
11	Spain

Cluster 2 — Moderate IMR (IMR per 1,000 live births)

#	Country		
1	Algeria	21	Marshall Islands
2	Bangladesh	22	Mauritania
3	Botswana	23	Micronesia, Fed. Sts.
4	Burkina Faso	24	Nepal
5	Cameroon	25	Palau
6	Congo, Rep.	26	Papua New Guinea
7	Dominica	27	Philippines
8	Dominican Republic	28	Rwanda
9	Egypt, Arab Rep.	29	South Africa
10	Eswatini	30	South Sudan
11	Fiji	31	Sri Lanka
12	Gabon	32	Tajikistan
13	Gambia, The	33	Tunisia
14	Ghana	34	Uganda
15	Guyana	35	United Kingdom
16	Haiti	36	Viet Nam
17	India	37	West Bank and Gaza
18	Iraq	38	Zambia
19	Kenya		
20	Kiribati		

Cluster 3 — Low IMR (IMR per 1,000 live births)

#	Country		
1	Antigua and Barbuda	27	Guatemala
2	Australia	28	Honduras
3	Austria	29	Hungary
4	Bahrain	30	Iceland
5	Barbados	31	Indonesia
6	Belarus	32	Ireland
7	Belgium	33	Israel
8	Bolivia	34	Italy
9	Brazil	35	Jamaica
10	Brunei Darussalam	36	Japan
11	Bulgaria	37	Jordan
12	Cabo Verde	38	Korea, Rep.
13	Canada	39	Kuwait
14	Chile	40	Latvia
15	Colombia	41	Lebanon
16	Costa Rica	42	Libya
17	Cuba	43	Lithuania
18	Czechia	44	Luxembourg
19	Denmark	45	Malaysia
20	Ecuador	46	Malta
21	El Salvador	47	Mauritius
22	Estonia	48	Mexico
23	Finland	49	Morocco
24	France	50	Nauru
25	Greece	51	Netherlands
26	Grenada	52	New Zealand
		53	Nicaragua

54	Norway	81	Switzerland
55	Oman	82	Tanzania
56	Panama	83	Thailand
57	Paraguay	84	Timor-Leste
58	Peru	85	Togo
59	Poland	86	Tonga
60	Portugal	87	Trinidad and Tobago
61	Qatar	88	Turkiye
62	Romania	89	Turkmenistan
63	Russian Federation	90	Tuvalu
64	Samoa	91	Ukraine
65	San Marino	92	United Arab Emirates
66	Sao Tome and Principe	93	United States
67	Saudi Arabia	94	Uruguay
68	Senegal	95	Uzbekistan
69	Seychelles	96	Vanuatu
70	Sierra Leone	97	Venezuela, RB
71	Singapore	98	World
72	Slovenia	99	Yemen, Rep.
73	Solomon Islands	100	Zimbabwe
74	Somalia, Fed. Rep.		
75	South Asia		
76	St. Lucia		
77	St. Vincent and the Grenadines		
78	Sudan		
79	Suriname		
80	Sweden		

10. Interpretation of Clusters

Cluster 1 — High IMR

The mean IMR of Cluster 1 was **63.89 in 2020**, which is already very high. It further **increased to 65.03 in 2021** and peaked sharply at **76.05 in 2022**, suggesting a major crisis or shock event (likely driven by Central African Republic's extreme spike). After that, it dropped to 59.95 in 2023 and 58.65 in 2024. Despite the decline, the values remain dangerously high, showing that these countries still face severe infant mortality burden with no consistent improvement.

Cluster 2 — Moderate IMR

Cluster 2 shows a **steady and consistent decline** from 32.43 in 2020 to 29.68 in 2024. This gradual downward trend suggests that these countries are making **slow but real progress** in reducing infant deaths, possibly through improving healthcare access, vaccination coverage, and maternal health programmes. However, the values are still far from acceptable levels. India comes under this cluster.

Cluster 3 — Low IMR

Cluster 3 has the lowest mean IMR, starting at **8.34 in 2020** and gently declining to **7.71 in 2024**. The values are consistently low and the decline is smooth, reflecting **well-established healthcare systems** and stable living conditions in these countries. These nations have largely achieved the goal of low infant mortality and are maintaining it steadily over time.

Overall summary

The three clusters reveal a clear stratification in global infant mortality. Cluster 1 nations carry the heaviest burden with high and volatile IMR, Cluster 2 nations are in a transitional phase showing gradual improvement, and Cluster 3 nations have achieved consistently low IMR. This pattern highlights deep inequalities in healthcare infrastructure and child survival outcomes across the world.

References

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3. **Anderson, T. W. (1971).** *The Statistical Analysis of Time Series*. New York: John Wiley & Sons.
4. **Anderson, T. W. (1971).** *The Statistical Analysis of Time Series*. New York: John Wiley & Sons.
5. World Bank Open Data — Infant Mortality Rate:
<https://data.worldbank.org/indicator/SP.DYN.IMRT.IN?locations=IN>

Appendix I: R Code — Time Series Analysis (India, 1971–2024).

The following R code was used to perform the complete time series analysis: data loading, stationarity testing, differencing, ARIMA model identification and fitting, diagnostics, and forecasting.

```
library(tseries)
data1=read.csv("C:/Users/sneha/Downloads/IMR.csv");data1
summary(data1)
str(data1)
ts_data=ts(data1[,2],c(1971,1));ts_data
plot.ts(ts_data)
adf.test(ts_data)
kpss.test(ts_data)
diff1=diff(ts_data,differences = 1);diff1
plot(diff1,main="first difference")
adf.test(diff1)
kpss.test(diff1)
diff2=diff(ts_data,differences = 2);diff2
plot(diff2,main="second difference")
adf.test(diff2)
kpss.test(diff2)
acf(diff2)
pacf(diff2)
model1=arima(ts_data,order=c(2,2,1));model1
model2=arima(ts_data,order=c(3,2,1));model2
model3=arima(ts_data,order=c(4,2,1));model3
model4=arima(ts_data,order=c(3,2,0));model4
model5=arima(ts_data,order=c(2,2,1));model5
model6=arima(ts_data,order=c(2,2,2));model6
model_list=list(model1,model2,model3,model4,model5,model6);model_list
```

```

aic_values=sapply(model_list,AIC);aic_values
bic_values=sapply(model_list,BIC);bic_values
arimamodel=arima(ts_data,order=c(2,2,2));arimamodel
Box.test(arimamodel$residuals,lag=10,type="Ljung-Box")
hist(residuals(arimamodel),
     probability = TRUE,
     main = "Histogram of Residuals",
     xlab = "Residuals")

lines(density(residuals(arimamodel)), lwd=2)

curve(dnorm(x,
     mean=mean(residuals(arimamodel)),
     sd=sd(residuals(arimamodel))),
     add=TRUE,
     lty=2,
     lwd=2)
install.packages("forecast")
library(forecast)
forecast_values <- forecast(arimamodel, h = 10);forecast_values
plot(forecast_values,main = "10-Year Forecast")

```

Appendix II: R Code — Cluster Analysis (149 Countries, 2020–2024).

The following R code was used to perform the complete cluster analysis: Elbow Method, Hierarchical Clustering using Ward's linkage, and K-Means Clustering with cluster assignment and interpretation.

```
#cluster analysis
# Scree Plot
install.packages("readxl")
library(readxl)
data1=read_excel("C:/Users/sneha/Downloads/imrdata.xlsx");data1
View(data1)
data=data1[,-1];data
str(data)
wss=(nrow(data)-1)*sum(apply(data,2,var));wss
for(i in 2:10){
  wss[i]=sum(kmeans(data,center=i)$withinss)
}
wss[i]
plot(1:10,wss,type="b",xlab="No. of clusters",ylab="within groups sum of squares")

# Hierarchical Clustering (Ward's Method)

data2=data;data
d=dist(data2,method="euclidean")
fit=hclust(d,method="ward.D")
plot(fit)
groups=cutree(fit,k=3)
groups
rect.hclust(fit,k=3,border="red")

#K-means clustering
fit1=kmeans(data2,3);fit1
aggregate(data,by=list(fit1$cluster),FUN=mean);
mydata=data.frame(data2,fit1$cluster);mydata
```